UNIVERSITY OF CALIFORNIA BERKELEY
Haas School of Business

MBA 201A — Economic Analysis for Business Decisions
Fall 2018

Professor Steve Tadelis

Final EXAM Blue and Gold

Instructions:

Please write your name and sign the honor code oath <u>on the last page</u> of this exam before starting the exam. Do this now so you don't forget later.

The number in brackets (*e.g.,* [5]) indicates the points for each part of each question. The total points per question are listed in the question header. Total: 155 points. Note that you have 180 minutes to do the exam, so if you spend about 1 minute per point, you will have 25 minutes to check your work.

Please Write Legibly. Briefly explain your answers (don't just write "yes" or "no" and don't just write down a numerical answer without showing how you derived it).

GOOD LUCK!

(For what it's worth, luck is always a component…)

<u> Please sign the honor code oath at the bottom of this page. ***</u>***

****

Honor Code:

I understand that this exam is an individual exercise. I swear on my honor that I have not consulted with another person or made use of notes or other materials on- or off-line during the exam.

Name:_____

Cohort:_____

Short answer questions [20]

The following **five** questions require only short answers (3-5 sentences). Use any graphs and economic reasoning that will help your explanation. Be sure to label graphs clearly.

1. [5] Last week Amazon increased the minimum wage of its workforce to $15 an hour. How do you think that this will affect the supply of workers and wages at Walmart in places where Amazon has fulfillment centers? How do you think this will affect the supply of workers and wages at in Seattle, where Amazon and Microsoft compete over employees?

2. [5] Consider companies that use low-skilled labor near an Amazon fulfillment center that were just about indifferent between hiring labor and switching to more automated robotic systems. How will Amazon's wage increase affect their production decisions? Do you think this may have an impact in the market for robotic devices?

3. [5] Tickets to a 49er's game are sold out and available only on StubHub. Sally and Mike are big 49er's fans. Mike tells Sally that he will buy some tickets and asks her how many she wants. Sally replies: "If the price per ticket is $350, buy one ticket for me. If the price per ticket is $250, buy two tickets for me. And, if the price per ticket is $150, buy three tickets for me." Mike responds: "You're saying you're willing to pay more in total for two tickets than for three. You are behaving irrationally." Sally replies, "I am totally rational. You should take an economics class." Who is right and why?

4. [5] A friend who heard your studying economics presents you with data on price and quantity in the market for Indium, a metal used in the construction of flat panel displays, and is traded on a worldwide market with many buyers and sellers. Your friend states that the demand curve for Indium must slope upwards—the higher the price, the more Indium consumers are willing to buy. To support the argument, your friend notes that, between the year 2015 and the year 2018, the price of Indium rose from roughly $180/Kg. to over $300/Kg, while the worldwide production of Indium rose from under 300 tons to over 500 tons. Do you think your colleague is correct about slope of the demand curve for Indium? Explain your argument in favor or against your colleague's statement.

Question 5: Tulsa Automotive [45 points]

Back in 2013 Tesla departed from industry practice and decided to sell cars directly to the public rather than sell through a network of dealers. Consider the following situation regarding a car manufacturer's choice of sales channel. Tulsa Automotive Co. (TAC), a new car manufacturer in Oklahoma, has developed a hydrogen cell fueled car, called the HX1, which it is now preparing to sell to the public. The cost of manufacturing each HX1 vehicle is $20,000 and TAC estimates it has a manufacturing capacity of 500 vehicles per year. In addition to per-vehicle manufacturing costs, TAC has annual overhead costs of $7.5 million, of which marketing and sales costs constitute $3.5 million. Tulsa estimated that yearly demand for the HX1 can be approximated by the following demand function:

$$q = 2{,}500 - 0.05p$$

where q is the quantity demanded and p is the price that TAC charges.

a. [5] What price should TAC charge? How much will it sell, and what will total profits be?

Suppose that additional market research reveals that much of the demand for the HX1 is concentrated in California. The company estimates that demand in CA can be approximated by the following function:

$$q_{CA} = 2{,}000 - 0.04p$$

While the rest of the U.S. market is the remainder from what was estimated earlier. TAC believes that by requiring buyers to show a driver's license issued at least 1 year prior to the purchase date, it can effectively provide different prices in California than in the rest of the U.S. Assume that TAC's plan to sell the HX1 at different prices in California than in the rest of the US would work, and to do this they would need to spend another $20,000 a year on personnel to verify residency. (For what follows, use q_R to denote the quantity of the rest-of-the-U.S. and use p_{CA} and p_R for prices respectively.)

b. [15] What should TAC do to maximize profits? What will profits be?

Instead of selling directly to the public, TAC can sell through a national company that owns a network of car dealers. Using the dealer network would negate the ability for the HX1 to be sold at different prices in California than in the rest of the country so the demand curve for the HX1 by consumers is the same is in part a). The national dealer company, a monopoly of dealers, knows the national demand. The upside of selling via dealers is that TAC would save 100% of the marketing and sales costs given above, regardless of the number of cars sold. Dealers buy the HX1 from TAC at a wholesale price and do not incur any additional costs beyond buying the HX1 cars from TAC. However, the dealers have the right to set the national retail price on cars sold to the public.

c. [10] Suppose that TAC decides to sell through the national dealer network and charges them a wholesale price equal to your solution from part (a). Put yourself in the position of the manager of the dealer network. How many cars would you order from TAC at that wholesale price, and how much higher would you set the retail price to sell the HX1 to the public?

d. [15] Now put yourself in the position of TAC. Given how the dealer network will mark up retail prices, at what wholesale price would you recommend TAC sell the HX1 to dealers? Please provide a rationale for your recommendation. What are your predicted profits for TAC and would you recommend they sell cars through the dealers or sell directly to the public?

Question 6: Product Lines [30 points]

A company is looking to sell its product in two versions, Standard and Luxury. Producing the standard version costs **$5** per unit and producing the luxury version costs **$30** per unit. You can assume that all other costs are sunk. Market research shows that there are **two types** of consumers interested in buying the products: **high spending** and **low spending**. There are **200** low spending consumers and **500** high spending consumers. The willingness to pay (WTP) for the standard and luxury versions of the product for each consumer group are reported in the table below. Consumers will buy at most one unit of the product.

	Standard	Luxury
High Spending	$90	$110
Low Spending	$60	$90

a. [5] If the company was ONLY considering selling the Luxury version, what price would it set and how many would it sell? What would its profits be?

b. [5] If the company was ONLY considering selling the Standard version, what price would it set and how many would it sell? What would its profits be?

c. [20] Would the company want to sell both products? If so, how much would they charge for each product?

Question 7: Ad Campaigns [30 points]

As the midterm elections are coming up, each one of two political candidates, 1 and 2, can choose to buy time on commercial radio shows to broadcast negative ad campaigns against their rival. These choices are made simultaneously. Denote the ad campaign time chosen by player i to be a_i, where i can be 1 or 2. Given a pair of choices (a_1, a_2), the payoff of candidates 1 and 2 are symmetric, and are given by the following functions:

$$v_1(a_1, a_2) = a_1 - 2a_2 + a_1 a_2 - {a_1}^2$$

$$v_2(a_1, a_2) = a_2 - 2a_1 + a_1 a_2 - {a_2}^2$$

a) [5] Imagine that player 1 believes that player 2 will choose $a_2 = 3$. What is player 1's best response?

b) [5] Does either candidate have a dominant action/strategy? What about a dominated one?

c) [10] What is the Nash Equilibrium of this game? Why is this the Nash Equilibrium?

Now that Kavanaugh joined the supreme court, and has demonstrated his strong ethics for not engaging in partisanship, he is suggesting that the Supreme court rule that no negative ad campaigns should be allowed.

d) [10] Do you think the two politicians would like the court to rule in favor of Kavanaugh's suggestion?

Question 8: IT Services [30 points]

Your company offers highly sophisticated information technology consulting services to clients in the retail industry. Serving these clients requires consulting time and computer resources. You have 50 consultants on staff. Each consultant costs you $10,000 per month in salary and benefits. Over the time frame relevant for this problem, you cannot hire more consultants. You can, however, reduce the number of consultants on staff. For each consultant you dismiss, you can save $10,000 per month. As usual, you seek to maximize profits.

You can serve two types of clients: "Big" and "Small." You are able to distinguish Big clients from Small clients. Each job for a Small client requires one month of time for one of your consultants, plus $2,000 worth of computer resources, which you purchase from Amazon Web Services. Each job for a Big client requires one month of time for one of your consultants, plus $5,000 worth of these same computer resources.

Small clients can meet their needs using services offered by many other consulting firms. These firms do not come close to your level of sophistication, but their availability puts an upper limit of $15,000 per job on what Small clients will pay for your services. You can get as many jobs from Small clients as you want at $15,000 per job, but none will pay more than $15,000 per job.

Big clients have more demanding needs. They cannot turn to the less sophisticated consultants. Big clients differ in their willingness to pay for your unique consulting services. The demand for your services by Big clients is given by

$P_B = 30{,}000 - 300Q_B$,

where P_B is the price you charge per job and Q_B is the number of jobs per month you can get from Big clients.

a) [15] What price per job should you set for Big clients? How many jobs per month will you handle for Big clients at that price? How many jobs per month will you handle for Small clients? What are your monthly profits?

b) [5 points] Is arbitrage a threat to your pricing strategy here? Explain why or why not.

c) [10] Suppose that a new open source software tool allows the many other consulting firms to better serve Small clients, driving down the price you can get from Small clients to $10,000 per job. (You do not use this tool, so your costs and the quality of your service are unchanged.) Now what price per job should you set for Big clients, and how many jobs per month will you handle for Big clients? How many jobs per month will you handle for Small clients? What are your monthly profits? Compare your answer in this part to your answer in part (a) and explain any differences.